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ZoomPursuit: Smooth Pursuit-based Gaze Interaction for General-Purpose GUIs

Master's Thesis

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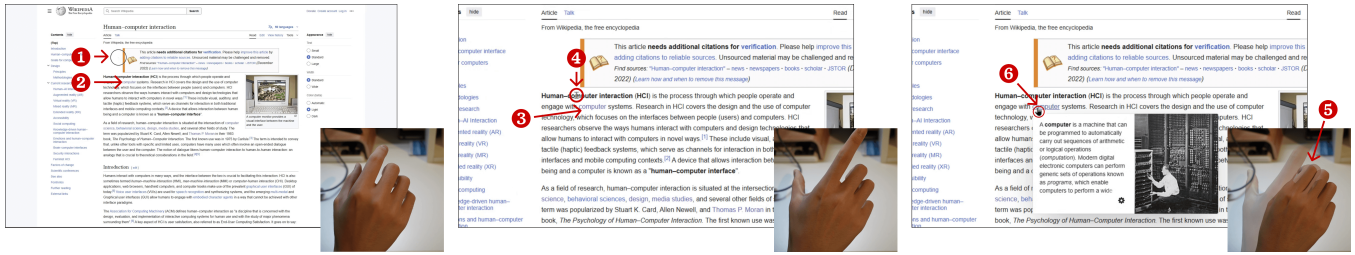


Figure 1: The two-stage ZoomPursuit workflow for high-accuracy smooth pursuit interaction on unmodified GUIs. Stage 1 (ZoomPursuit Automated Correction): A user fixates on a target (❶) while the system measures an inaccurate gaze point (❷, gaze indicator). A raised finger triggers a zoom centered on ❶, displacing the target from ❷ to ❸. This prompts a natural smooth pursuit eye movement as the user tracks the moving target. The system analyzes this pursuit to automatically correct the pointer to ❸. **Stage 2 (Manual Refinement):** ❹ The user performs optional, fine-grained adjustments with hand movements. ❺ A final pinch gesture selects the accurately positioned target.

Abstract

Gaze interaction in general-purpose GUIs is limited by its inaccuracy and calibration requirements. Smooth pursuit-based methods offer a calibration-free alternative but often demand modified user interfaces. We present ZoomPursuit, a technique that adapts smooth pursuit for unmodified GUIs by repurposing system-level magnification functionality.

A baseline evaluation ($N=16$) established that under simulated calibration drift, ZoomPursuit achieved the accuracy of a manual refinement method with the speed of a simpler automated zoom-based technique. However, the study also revealed a conflict between its imperceptible automation and users’ sense of control. A follow-up study ($N=7$) resolved this with a two-stage model integrating automated correction with manual refinement. This design achieved a final pointing error of 0.10° while dynamically calibrating on the fly from an uncalibrated state.

This work details the design evolution of a pursuit-based interaction technique for general GUIs. The final design demonstrates a model that is both technically effective and preserves user agency.

1 Introduction

Gaze interaction offers a fast, intuitive pointing method, as a user’s gaze often reflects their intent [Duchowski 2018; Shakil et al. 2025]. However, its adoption in general-purpose Graphical User Interfaces (GUIs) is hindered because GUIs demand a higher level of pointing accuracy than the human eye can naturally provide [Feit et al. 2017; Fernandes et al. 2024]. This ocular “fat finger” problem arises because the fovea, the eye’s area of sharpest vision, spans approximately 1° , making it less precise than pixel-level input devices like a mouse [Bates and Istance 2002]. Two technical factors compound this limitation. First, eye tracking hardware introduces measurement errors that reduce accuracy [Feit et al. 2017]. Second, calibration drift, caused by changes in user posture or device position, degrades performance over time and forces disruptive recalibrations [Gomez and Gellersen 2018; Sugano and Bulling 2015]. The need for per-user calibration also hinders immediate, casual

use [Sugano and Bulling 2015]. Furthermore, dynamically correcting for drift is difficult because everyday interactions rarely provide the unambiguous gaze-target data needed for effective recalibration.

One approach to address these issues uses the relative motion of smooth pursuit eye movements for pointing, which can operate without an explicit calibration phase [Pfeuffer et al. 2013; Vidal et al. 2013]. By matching the eye’s motion pattern to that of an on-screen target, these techniques can mitigate errors from absolute gaze coordinates. However, this approach typically sacrifices generality. Most pursuit-based techniques require modifying the GUI to embed dynamic visual elements for the user to follow [Esteves et al. 2015; Khamis et al. 2018; Zeng et al. 2020]. This requirement makes them incompatible with standard, unmodified interfaces, limiting their use as a general pointing method. This limitation defines a key challenge for smooth pursuit interaction research [Shakil et al. 2025]: **how can we design a smooth pursuit technique that retains smooth pursuit’s benefits and is applicable to general-purpose GUIs?**

We address this challenge with **ZoomPursuit** (see the first two steps in Figure 1), a technique that generates the required motion stimulus by repurposing a standard system-level magnification function. When a user triggers the interaction, the system executes a smooth zoom centered on their initial, and potentially inaccurate, gaze point. This magnification displaces the user’s intended target, inducing a natural smooth pursuit eye movement. The underlying principle is that this zoom-induced motion creates a unique vector for every point on the screen relative to the zoom’s center. By analyzing the user’s pursuit, the system captures this vector and infers the pointing error from relative motion, not absolute coordinates.

We present the design and evaluation of this technique through two sequential user studies. The first is a baseline evaluation that establishes the technique’s performance under controlled error conditions. Informed by these findings, a follow-up study evaluates an integrated two-stage workflow designed to address the limitations of a fully automated approach and demonstrate the technique’s viability in a more challenging, uncalibrated scenario.

2 Related Work

This section reviews prior work that addresses three barriers to gaze interaction: initial calibration overhead, inherent inaccuracy, and calibration drift.

2.1 Zoom-based Pointing

One approach to mitigate pointing error is to magnify the area around the user’s gaze point [Hansen et al. 2008; Stellmach and Dachselt 2012]. This technique lowers the required pointing accuracy, making it easier to select small targets. This principle is used in commercial systems like TD Control¹, which employs a two-step zoom, and in research prototypes such as EyePoint [Kumar et al. 2007]. Windows Eye Control² also implements this, but requires users to fine-tune their selection by gazing at on-screen directional controls after magnification.

However, these methods only reduce the effect of pointing errors, rather than correcting the underlying error source. The systematic offset from eye tracking error and calibration drift remains. This presents a trade-off for the user: a high magnification factor can manage a large error but may obscure the surrounding interface context. Alternatively, users might learn to compensate for the inaccuracy by looking away from the intended target to place the cursor correctly, which increases cognitive load. If the error becomes too large, users must interrupt their workflow to perform a full recalibration. Gaze data collected during interaction is potentially biased by these compensatory looking strategies, making it an unreliable data source for dynamically correcting calibration drift. This behavior creates a data quality issue, as the system receives gaze points that reflect a workaround, not the user’s true intent, hindering its ability to learn a correct error model.

2.2 Multimodal Refinement

Multimodal refinement combines an initial gaze selection with a pointing action from a more precise and accurate secondary input, such as a mouse [Zhai et al. 1999], touch [Stellmach and Dachselt 2013], or head tracking [Kytö et al. 2018]. A systematic evaluation in an AR context found that a secondary input, like a hand gesture or head motion, could reduce gaze pointing error from over 2° to a range of 0.3° to 0.5° [Kytö et al. 2018]. The secondary input is typically more precise and is not affected by gaze-specific issues like calibration drift.

However, this reliance on a non-gaze modality limits its use in purely gaze-based contexts, a key requirement for some assistive technologies. The study also suggested that consistent offsets recorded during refinement could correct for calibration drift in real-time [Kytö et al. 2018], a concept that informs the implicit calibration techniques discussed next.

2.3 Implicit Calibration

Previous approaches either mitigate an error’s effects or rely on external inputs. Implicit calibration techniques correct the gaze signal itself by dynamically counteracting drift during natural use [Morimoto and Mimica 2005; Shakil et al. 2025]. These methods work

by collecting high-confidence “gaze-target” data pairs during interaction to infer pointing error.

However, strategies to collect these data pairs have key limitations. Methods relying on clicks on known UI elements compromise generality, as they require system knowledge of the UI and fail in content-heavy areas [Hornof and Halverson 2002; Sidenmark and Lundström 2019]. Other approaches leverage visual saliency models to predict user focus during natural viewing [Sugano and Bulling 2015], but their effectiveness can be limited as they must passively wait for suitable gaze-target data to emerge, which may not occur frequently enough to prevent calibration drift.

2.4 Pointing with Smooth Pursuit

Smooth pursuit interaction leverages the human ability to follow a moving object for pointing and selection [Vidal et al. 2013]. The underlying principle is that by matching a user’s gaze path to a target’s known trajectory, the system can determine the intended selection. This process also measures the gaze offset, which allows the technique to be adapted for on-the-fly dynamic calibration [Gomez and Gellersen 2018; Murauer et al. 2018; Pfeuffer et al. 2013; Santini et al. 2017].

This capability typically comes at the cost of generality. Most pursuit-based solutions require modifying the target application to embed motion stimuli. This is because these techniques traditionally rely on matching the user’s eye movements to a target’s **pre-defined and unique trajectory**, which is most easily created by altering the visual elements of the UI. As Figure 2 illustrates, these modifications range from moving the target itself to adding orbiting or tracing motions. This modification requirement is often infeasible for general-purpose GUIs. To the best of our knowledge, SPOCK [Schenk et al. 2017] is the only pursuit-based system designed for general-purpose GUIs [Shakil et al. 2025], but its mechanism of projecting an external stimulus was aimed at addressing the Midas Touch problem, not improving pointing accuracy. This leaves a gap for a method that can provide active, pursuit-based error correction without requiring changes to the underlying application.

3 The ZoomPursuit Technique

This section presents the technique’s design goals, core mechanism, and performance model.

3.1 Design Goals

Five goals guided the design. For a gaze pointing technique to be viable on mouse-centric desktop GUIs, it must: (1) **Achieve High Accuracy**, (2) **Operate on Unmodified Applications**, (3) **Correct Pointing Errors** during each interaction to counter the inherent inaccuracy of the gaze signal, (4) **Enable Dynamic On-the-fly Calibration** to combat long-term calibration drift, and (5) **Support Gaze-only Operation** for broad applicability.

3.2 Workflow and Core Mechanism

To understand and correct for pointing error, we first formally model it. We model the pointing error as a position-dependent error field, $E(P)$, where the measured gaze point $G_{measured}$ relates to the true gaze point G_{true} by $G_{measured} = G_{true} + E(G_{true})$.

¹<https://www.tobiidynavox.com/products/td-control>

²<https://support.microsoft.com/en-us/windows/eye-control-basics-in-windows-97d68837-b993-8462-1f9d-3c957117b1cf>

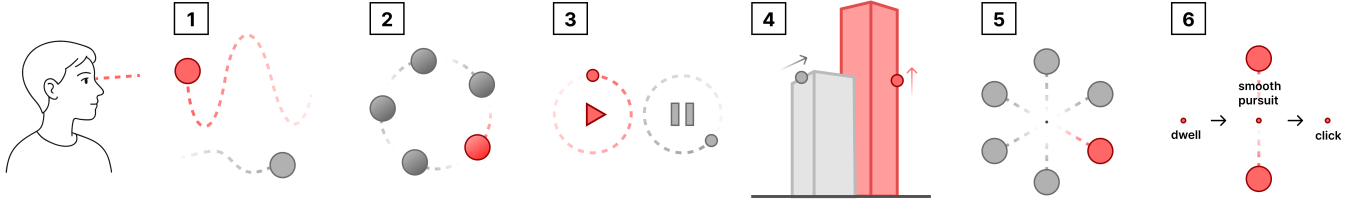


Figure 2: Examples of prior smooth pursuit techniques that use different motion stimuli to enable interaction. These include (1) moving the target along a preset path (Pursuits [Vidal et al. 2013]), (2) arranging elements in a 3D pattern (VRPursuits [Khamis et al. 2018]), (3) adding orbiting motion stimuli around a button (Orbits [Esteves et al. 2015]), (4) adding motion stimuli around an object’s outline (Outline Pursuit [Sidenmark et al. 2020]), (5) radiating targets outwards (Linear Smooth Pursuit [Zeng et al. 2020]), and (6) injecting a stimulus via reflection (SPOCK [Schenk et al. 2017]). Images are redrawn from the cited publications.

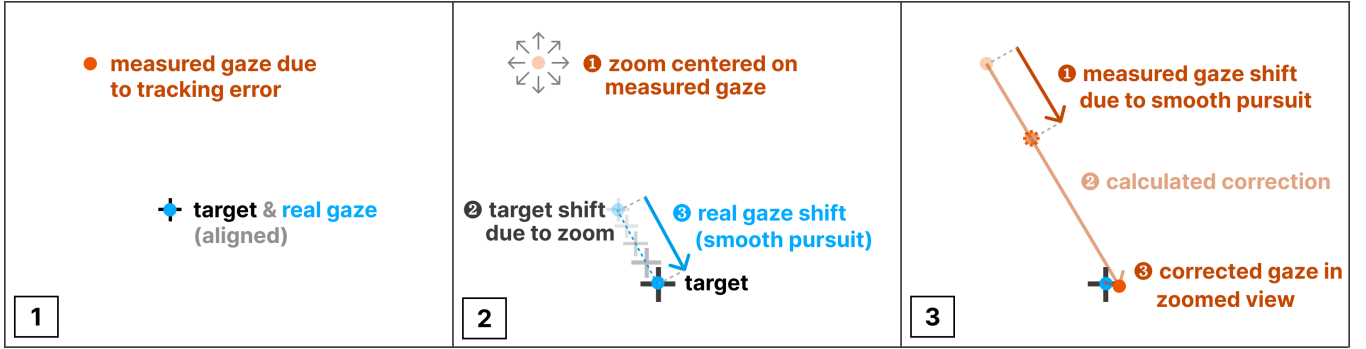


Figure 3: The workflow of the ZoomPursuit technique. (1) Initially, a pointing error, $E_{initial}$, exists between the true target position, $G_{true,initial}$, and the measured gaze point, $G_{measured,initial}$. (2) The view is magnified, centered on the measured gaze point. This displaces the true target, inducing a pursuit eye movement. (3) The system analyzes the resulting gaze shift, $\Delta G_{measured}$, to calculate an estimate of the initial error, $E_{estimated}$, and correct the pointer’s position.

The interaction begins when a user signals an intent to interact, as Figure 3 illustrates. The system magnifies the screen, centered on the initial, erroneous gaze point, $G_{measured,initial}$ (Figure 3, step 2). This action displaces the true target, $G_{true,initial}$, on the screen. The target’s displacement, ΔP_{target} , is proportional to the initial error, $E_{initial}$, and the change in zoom scale:

$$\Delta P_{target} = (S_{end} - S_{start}) \cdot E_{initial} \quad (1)$$

This displacement induces a smooth pursuit eye movement. The correction algorithm’s core is to analyze the resulting gaze shift, $\Delta G_{measured}$, to estimate the initial error, $E_{initial}$, and correct the pointer’s position (Figure 3, step 3). This is achieved by inverting Equation (1):

$$E_{estimated} = \frac{\Delta G_{measured}}{S_{end} - S_{start}} \quad (2)$$

To analyze this approach’s performance boundaries, we must consider that Equation (2)’s estimation is affected by two main factors: the local gradient of the error field, $\Delta E = E(G_{true,final}) - E(G_{true,initial})$, and the eye tracker’s signal noise, $\epsilon_{pursuit}$. This leads to the final performance model:

$$E_{estimated} \approx E_{initial} + \frac{\Delta E}{S_{end} - S_{start}} + \frac{\epsilon_{pursuit}}{S_{end} - S_{start}} \quad (3)$$

The correction mechanism’s validity hinges on a critical assumption: the user’s pursuit must accurately follow the target’s displacement.

3.3 Performance Analysis and Design Implications

The performance model (Equation 3) shows that the technique transforms correcting a large absolute error ($E_{initial}$) into mitigating two smaller residual errors. This conversion from an absolute to a relative pointing task is central to achieving high accuracy. The first residual term, related to the error field’s local gradient (ΔE), diminishes if the field is locally smooth. The second term, from eye tracker signal noise ($\epsilon_{pursuit}$), is reduced when the signal is precise.

This analysis has direct design implications that informed our two-part evaluation strategy.

First, it was necessary to assess the performance of the core ZoomPursuit mechanism under controlled conditions to understand its fundamental characteristics. **Our baseline evaluation was designed for this purpose**, systematically varying the initial pointing error to evaluate its effect on the automated correction.

Second, the performance model suggests that in scenarios with large, unpredictable error gradients, such as a fully uncalibrated system, the residual errors might remain significant. This insight motivated **our follow-up study, which explores a two-stage**

design that integrates manual refinement to ensure final accuracy in more demanding contexts.

Finally, the correction algorithm (Equation 2) does not require continuous pursuit tracking. Because it relies only on the stable start and end points of the gaze trajectory, it is robust to dynamic tracking imperfections like latency or gain errors.

4 Methodology

We evaluated ZoomPursuit through a two-part study. Study 1 was a baseline evaluation that established the technique’s performance in a controlled setting. Informed by its results, a follow-up study explored integrating the technique into a two-stage workflow for high-accuracy pointing.

4.1 Study 1: Baseline Evaluation

We assessed objective performance using pointing accuracy (final error) and speed (task time). We measured subjective experience using perceived workload (NASA-TLX) and usability (UMUX-Lite).

We compared ZoomPursuit against two baseline techniques, illustrated in Figure 4. **ZoomCenter**, modeled after systems like TD Control, magnifies a circular area around the user’s gaze point and presents this view in a static, centered window. The user then reacquires the target in this magnified view. **ZoomDPad**, modeled after Windows Eye Control, also magnifies the area around the gaze point. It then discretizes the user’s gaze into four directions, allowing them to move the magnified view in steps to center the target. We selected these methods because they represent established paradigms for pointing assistance.

4.1.1 Research Questions and Hypotheses. This led to our first research question and hypotheses.

RQ1: *How do ZoomPursuit and two baseline techniques (ZoomCenter, ZoomDPad) compare in terms of objective performance (accuracy, speed) and subjective experience (workload, usability) under accurate versus drifted calibration conditions?*

To simulate different levels of calibration quality, we developed a procedure to manipulate each participant’s pointing error in real-time. First, we measured each participant’s baseline error across a 5x4 grid of points (Figure 4, rightmost panel) to create a personalized error field. We modeled this field using Inverse Distance Weighting (IDW) interpolation based on the four nearest measured points.

An algorithm then manipulated this field with a personalized scaling factor to create two conditions. In the **Accurate** condition, we normalized the error to a mean of approximately 1°. In the **Drifted** condition, we amplified it to approximately 3°. This method simulated two distinct calibration quality levels: an adequate calibration (1°, a level often considered the upper bound for a well-calibrated consumer-grade eye tracker) and a realistic drift (3°), informed by prior work showing that real-world pointing errors often exceed 2° [Feit et al. 2017].

Our performance model led to the following hypotheses:

H1: *The accuracy of the absolute-coordinate technique (ZoomCenter) will degrade under calibration drift, while the relative-error (ZoomPursuit) and manual-refinement (ZoomDPad) techniques will remain robust.*

H2: *The more direct interactions of ZoomCenter and ZoomPursuit will result in faster task completion times compared to the multi-step refinement of ZoomDPad.*

H3: *Users will prefer ZoomPursuit and rate it as having a lower perceived workload compared to ZoomDPad and ZoomCenter.*

H4: *The final pointing accuracy of ZoomDPad will be independent of the initial error due to its explicit manual correction mechanism. In contrast, the final accuracy of ZoomPursuit and ZoomCenter will depend on the initial error.*

4.2 Study 2: Follow-up Study

This follow-up study explores a two-stage workflow combining gaze-based correction with manual refinement. This workflow had two aims: first, to address the user agency issues identified in Study 1; second, to explore how real-time error estimation could enable features like on-the-fly calibration. The first stage uses the ZoomPursuit mechanism to automatically correct initial pointing inaccuracies. The second stage allows optional, accurate and precise manual adjustments, using the initial error estimated in the first stage to set a dynamic step-length. This design led to a new set of research questions.

4.2.1 Research Questions. RQ2: *How accurate is the ZoomPursuit correction (first stage)?*

RQ3: *How accurate is the two-stage design?*

RQ4: *Can the two-stage refinement technique support on-the-fly calibration during interaction?*

RQ5: *What is the user experience of using the two-stage refinement design?*

To answer these questions, we conducted an exploratory, mixed-methods study. We developed a Gaze+Pinch prototype integrating ZoomPursuit with a manual refinement stage. We collected quantitative data for **RQ2**, **RQ3**, and **RQ4**, and qualitative data for **RQ5**.

5 Study 1: Baseline Evaluation of Automated Correction

We conducted a controlled experiment with two primary goals. First, as a baseline evaluation, we assessed the pointing performance of ZoomPursuit against representative alternative techniques. Second, we used this evaluation to understand the boundaries of this automated approach, particularly regarding user experience.

5.1 Participants

We recruited 16 university participants, aged 18 to 34 ($M=23.1$, $SD=3.8$). The group included 7 males, 7 females, and 2 who preferred not to specify their gender. Quantitative data for two participants were lost due to a technical issue where the online virtual disk allocated for the experiment became full; we therefore included 14 participants in the quantitative analysis but retained their qualitative data. All participants had normal or corrected-to-normal vision and received a 10 EUR e-gift card as compensation. Verbal communication was conducted in either English or Chinese, based on the participant’s preference. All written materials were in English.

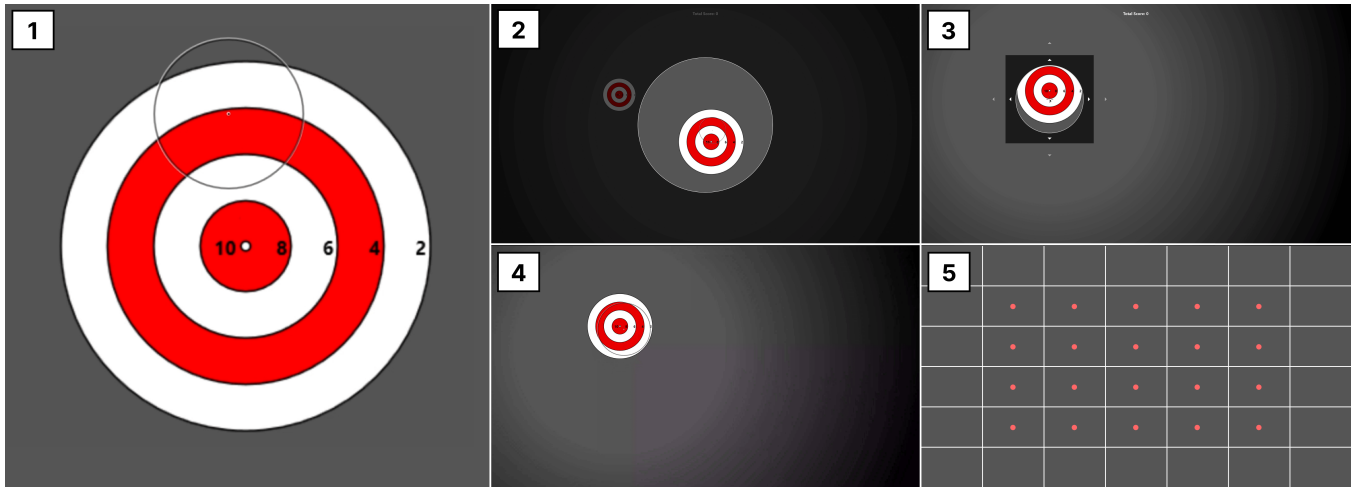


Figure 4: Elements of the Study 1 pointing task. (1) The bullseye target and gaze indicator. The figure shows the three interaction techniques at their initial 2x magnification: (2) *ZoomCenter*, (3) *ZoomDPad*, and (4) *ZoomPursuit*. (5) The task used 20 possible target locations, identical to those in the accuracy assessment. In all techniques, a user could cancel a trial by looking into the black surrounding area before releasing the spacebar.

5.2 Apparatus and Environment

We used a 24-inch 1080p monitor with a Smart Eye AI-X eye tracker (60Hz). Participants sat approximately 65 cm from the monitor without a chin rest; we used this distance to convert pixels to degrees of visual angle. An AutoHotkey script mapped the CapsLock key to a non-standard virtual key (vkFC). This approach treated the key as a modifier for the experiment, preventing interference with normal computer use (e.g., typing in questionnaires). A small weight was placed on the key to keep it continuously activated during the experimental blocks. This setup let participants use only the spacebar for all actions, minimizing extraneous cognitive load.

5.3 Experimental Design

We employed a 2x3 mixed-factorial design. The independent variables were **Interaction Technique**, a within-subjects factor with three levels (*ZoomPursuit*, *ZoomCenter*, and *ZoomDPad*), and **Calibration Condition**, a between-subjects factor with two levels (*Accurate* and *Drifted*). We randomly assigned participants to a calibration condition ($n=8$ per group). The order of the three techniques was counterbalanced for each participant using a balanced Latin square.

5.3.1 Interaction Techniques. We implemented all techniques in a single prototype using the Windows Presentation Foundation (WPF) framework. To ensure signal stability, we processed raw gaze data in real-time. Our approach followed the recommendations of a comprehensive filter review [Stuart et al. 2019]. First, a Velocity-Threshold Identification (I-VT) algorithm classified eye movements. Then, for an effective yet computationally efficient solution, a simple triangular filter smoothed the data. This pipeline produced a stable signal, using saccadic points for responsiveness and a smoothed fixation average to reduce noise. Figure 4 illustrates the target and visual feedback for each technique.

5.4 Task and Procedure

5.4.1 Task. Participants performed a standard pointing task. In each trial, a bullseye target (2° radius) appeared at one of 20 preset grid locations. These locations were identical to the points used for baseline accuracy sampling, allowing us to approximate the initial error for each trial. All techniques followed a unified “look-press-look-release” interaction flow [Kumar et al. 2007]. An initial key press triggered a 2x magnification. For *ZoomCenter* and *ZoomPursuit*, holding the key increased magnification to 4x, while *ZoomDPad* used a fixed 2x magnification, as its manual refinement stage already provided the means for accurate and precise pointing.

5.4.2 Procedure. The session began with a consent form and a demographic questionnaire (Appendix A.2). Next, we performed a 5-point calibration. Participants then completed a block for each of the three techniques in a counterbalanced order. Each block consisted of four steps:

- (1) **Accuracy Sampling:** An independent 20-point accuracy sampling collected baseline data. This data served two purposes: it was used to model a personalized error field for the experimental manipulation, and because the formal task used the same target locations, it provided an approximation of the initial pointing error for each trial.
- (2) **Practice:** Participants watched a tutorial video and completed six practice trials.
- (3) **Formal Task:** Participants completed 20 pointing trials. The experimenter could skip a trial if it became too difficult; these trials were marked as failed and excluded from analysis.
- (4) **Questionnaires:** Participants completed the NASA-TLX and Usability Metric for User Experience-Lite (UMUX-Lite) questionnaires after each task (Appendix A.1).

The session concluded with a semi-structured interview, which we audio-recorded (protocol in Appendix A.3.1). We logged all interaction data for analysis.

5.5 Data Analysis Strategy

All interaction data were logged in a detailed JSON-Lines (.jsonl) format. We used Python scripts to parse these logs and extract key metrics for analysis.

5.5.1 Model Selection and Specification. We used Linear Mixed-Effects Models (LMMs) to analyze the data. This approach is well-suited for our design, which nested multiple trials within each participant, and it allowed us to analyze each trial individually. Given our sample size ($N=14$), we included only random intercepts for participants. A more complex random effects structure (e.g., including random slopes) was not adopted because the small sample size offers limited statistical power to model such individual differences in condition effects, and simpler models reduce the risk of overfitting and convergence issues.

5.5.2 Assumption Checking and Data Transformation. We checked the LMM assumptions by inspecting the model residuals. This included using Quantile-Quantile (Q-Q) plots, a graphical method for comparing two probability distributions by plotting their quantiles against each other, to check for normality. When the data violated normality assumptions, we applied transformations (e.g., log or square-root) to meet the model's requirements.

5.5.3 Hypothesis Testing. For our manipulation check, we used an independent samples t-test, with the non-parametric Mann-Whitney U test as a fallback. To test our hypotheses, we used LMMs with post-hoc tests for significant effects. Our primary models analyzed pointing error, task time, and workload as functions of Technique, Condition, and their interaction. For **H4**, a separate LMM modeled final error as a function of initial error (approximated from the pre-task accuracy assessment at the same target locations) for each technique. Full model results are in Appendix A.4.

5.5.4 Qualitative Data Analysis. The researcher transcribed the interview recordings and used Thematic Analysis to develop a codebook and identify recurring themes in the qualitative data.

6 Study 1: Baseline Evaluation Results

This section presents the results of our baseline evaluation, reporting first on objective performance (pointing error, task time) and then on subjective findings from questionnaires and interviews.

We analyzed all dependent variables using Linear Mixed-Effects Models (LMMs). We applied a square-root transformation to pointing error data and a log-transformation to task time data to meet model assumptions. We conducted simple effects analyses for significant interactions. Full statistical output is available in Table 1 in the Appendix.

6.1 Manipulation Check

Our experimental manipulation of calibration drift was effective. The initial error data violated the t-test's normality assumption, so we used a Mann-Whitney U test. This test confirmed the median initial pointing error was significantly higher in the *Drifted* group

($Mdn = 2.32^\circ$) than in the *Accurate* group ($Mdn = 0.88^\circ$), $U = 49$, $p < 0.001$.

6.2 Trial Exclusions

A total of 6 out of 840 trials (0.7%) were excluded from the performance analysis. These exclusions comprised two types of events recorded during the experiment: trials that were manually skipped by the experimenter after a participant failed more than twice, and trials recorded as unsuccessful. The latter typically occurred due to edge cases in event detection, resulting in a final recorded error larger than the target's 2° radius. Given the small number of exclusions, this process is unlikely to bias the overall results.

6.3 Objective Performance

The model's fixed effects (technique, condition, interaction) accounted for 12.1% of the variance in pointing error ($R_m^2 = 0.121$) and 15.6% in task time ($R_m^2 = 0.156$). Including random intercepts for participants increased the explained variance to 25.3% for error ($R_c^2 = 0.253$) and 32.9% for time ($R_c^2 = 0.329$). Here, marginal R^2 (R_m^2) represents the variance explained by fixed effects alone, while conditional R^2 (R_c^2) represents the variance explained by both fixed and random effects. The difference between marginal and conditional R^2 values indicates individual differences accounted for a portion of the performance variation. Figure 5 shows the distributions of final pointing error and task time.

6.3.1 Pointing Error. To answer **RQ1** regarding pointing error, our LMM identified a significant interaction between technique and calibration condition ($p < 0.001$). We analyzed the simple effects for each condition to test **H1**.

Under drifted calibration, there was a significant effect of technique. Post-hoc comparisons showed *ZoomCenter* was significantly less accurate than *ZoomPursuit* ($\beta = 0.261$, $p < 0.001$), where β represents the estimated difference on the transformed error scale. This finding is consistent with **H1**. There was no significant accuracy difference between *ZoomDPad* and *ZoomPursuit* ($p = 0.405$).

Under accurate calibration, we found no significant difference between *ZoomPursuit* and *ZoomCenter* ($p = 0.133$). However, *ZoomDPad* was significantly more accurate than *ZoomPursuit* ($\beta = -0.102$, $p = 0.002$).

6.3.2 Task Completion Time. We also found a significant interaction effect for task time ($p < 0.001$).

Under drifted calibration, *ZoomDPad* was significantly slower than *ZoomPursuit* ($\beta = 0.643$, $p < 0.001$), a 90.2% increase in task time. There was no significant time difference between *ZoomCenter* and *ZoomPursuit* ($p = 0.396$).

Under accurate calibration, this pattern held. *ZoomDPad* remained significantly slower than *ZoomPursuit* ($\beta = 0.280$, $p < 0.001$), a 32.3% increase. There was no difference between *ZoomPursuit* and *ZoomCenter* ($p = 0.721$). These findings are consistent with **H2**.

6.4 Subjective Experience

We analyzed questionnaire and interview data to address the subjective component of **RQ1**.

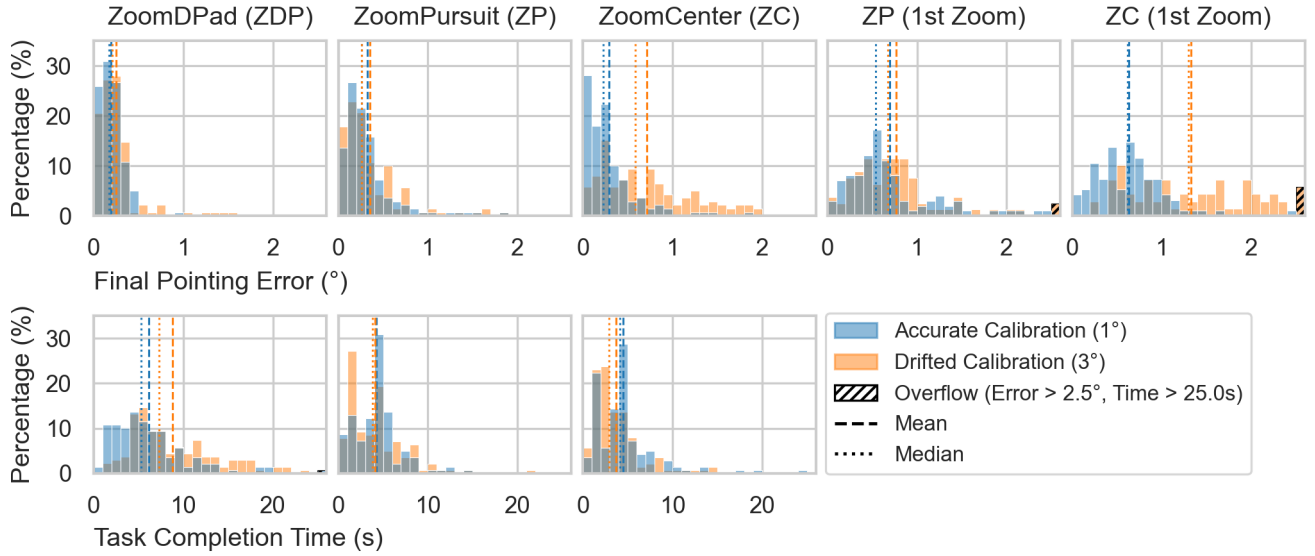


Figure 5: Distributions of final pointing error (top) and task completion time (bottom) for the baseline evaluation (Study 1). Results are grouped by interaction technique and calibration condition (*Accurate* vs. *Drifted*). The columns labeled “(1st Zoom)” show the pointing error measured after the initial automated magnification step. The data shows that while *ZoomCenter*’s accuracy degraded significantly under drift, *ZoomPursuit* and *ZoomDPad* remained robust. *ZoomDPad* was consistently the slowest technique.

6.4.1 Questionnaire Ratings: Workload and Usability. Our results did not support **H3**. We found a significant main effect of technique on NASA-TLX scores and a near-significant interaction ($p = 0.059$). Post-hoc tests showed that in the **Drifted** condition, *ZoomPursuit* had a significantly higher workload than *ZoomDPad* ($p = 0.020$). In the **Accurate** condition, there were no significant differences. We found no significant effects on UMUX-Lite usability scores after FDR (False Discovery Rate) correction.

6.5 Robustness to Initial Pointing Error

6.5.1 Thematic Analysis Findings. Our analysis of the interview data revealed four key findings.

Finding 1: Participants Prioritized a Sense of Control. This preference for predictable control was most evident in participants’ evaluation of *ZoomDPad*, which was often praised despite its slower pace. *ZoomDPad* was often cited for this quality, with one user stating it felt like “you are controlling the magnifying view, not that the gaze cursor is following you” (P3). Conversely, the lack of direct control in *ZoomCenter*, particularly under drift, led to varied coping mechanisms. Some in the *Drifted* group developed a compensatory strategy of looking away from the target (P4, P10). Others either abandoned the goal of achieving accuracy (P12) or found the method unusable without a workaround (P2, P8). Participants found little success in applying similar compensatory strategies to *ZoomPursuit*. Attempts to manually correct the pointer were often met with unstable and erratic behavior, which users described as “drifting” (P8), “floating a lot” (P13), or “always jumping around” (P14), making deliberate compensation ineffective.

Finding 2: Predictability Was Key to Forming a Mental Model. When a technique’s behavior was unpredictable, users struggled to form a coherent mental model and often expressed frustration. Users described the behavior of *ZoomCenter* and *ZoomPursuit* under drift with terms like “drifting” (P8), “jumping” (P14, P5), and “floating” (P13). In contrast, they described *ZoomDPad*’s mechanism as “orderly” (P8) and “predictable” (P4), which gave them a sense of agency. This suggests a trade-off: users often accepted a slower method if it provided a stable, understandable experience.

Finding 3: *ZoomDPad* Created a Conflict Between Means and End. While praised for its predictability, *ZoomDPad* introduced a usability challenge by forcing users to split their attention. This created a “conflict of means and end” (P11), where users had to fixate on the directional arrows while using peripheral vision to track the target (P15, P2). The four-direction mapping increased this cognitive load for some, making diagonal movements feel inefficient (P14). This decoupling of control was not universally seen as negative; one participant (P8) reported it let his brain “rest” between discrete control actions, unlike the constant high focus demanded by other techniques.

Finding 4: The *ZoomPursuit* Correction Mechanism Was Imperceptible. The automated correction at the core of *ZoomPursuit* was largely invisible to users, with most perceiving it as functionally identical to *ZoomCenter* (P3, P5). Consequently, when the technique failed, they attributed it to system instability (“it floated a lot,” P13) or found it ineffective (P1). This lack of understanding did not consistently lead to a negative perception. For several participants, the technique’s automated nature felt almost magical; it was

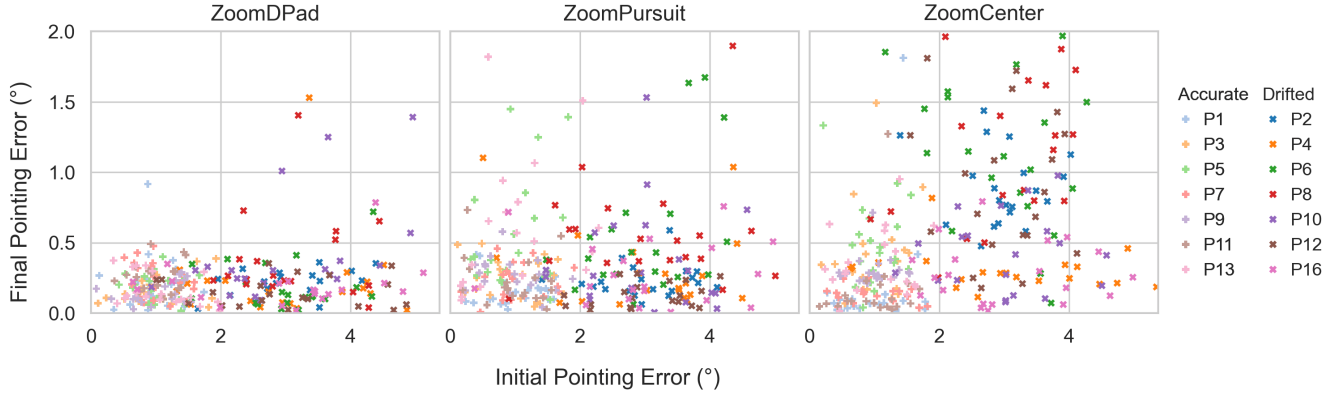


Figure 6: Final pointing error as a function of initial pointing error for each technique, across all trials from the baseline evaluation (Study 1). Each point represents a single trial.

a primary benefit, allowing them to achieve high accuracy with surprisingly low effort (P2, P12).

Preference Shift Across Conditions. Participants’ final preference rankings revealed a shift across calibration conditions. In the **Accurate** condition, 4/8 participants preferred *ZoomCenter*, 3/8 preferred *ZoomDpad*, and 1/8 preferred *ZoomPursuit*. In the **Drifted** condition, this pattern inverted: 4/8 participants preferred *ZoomPursuit*, 3/8 preferred *ZoomDpad*, and 1/8 preferred *ZoomCenter*.

To test **H4**, we modeled final pointing error as a function of initial error for each technique (Figure 6). The results did not support our hypothesis. As predicted, initial error did not significantly predict final error for *ZoomDpad* ($p = 0.081$). However, contrary to our prediction, it was also not a significant predictor for *ZoomPursuit* ($p = 0.321$) or *ZoomCenter* ($p = 0.419$). We address this finding further in the Discussion section.

7 Study 2: Follow-up Study

Our baseline evaluation (Study 1) established *ZoomPursuit*’s objective performance but also identified a conflict between its automation and users’ subjective experience, related to a perceived lack of transparency and control. This finding, combined with our theoretical model’s prediction that uncalibrated scenarios require additional support, motivated the design of a two-stage interaction model. This follow-up study evaluates the resulting two-stage interaction model.

This study evaluates the two-stage interaction model, addressing the system’s automated correction (**RQ2**), overall pointing performance (**RQ3**), on-the-fly calibration (**RQ4**), and user experience (**RQ5**).

7.1 Participants

We recruited 7 researchers (1 aged 18-24, 6 aged 25-34; $M=28.3$, $SD=3.2$; 5 male, 1 female, 1 non-binary). The small sample size resulted from limited access to the required eye-tracking hardware (Tobii Pro Fusion), as its API allows for acquiring gaze data even without a prior user-specific calibration, a feature essential for our

dynamic calibration model. All participants had normal or corrected-to-normal vision and received a 10 EUR e-gift card.

7.2 Apparatus and Environment

We used a 27-inch 1440p monitor with a 120Hz Tobii Pro Fusion eye tracker. Participants sat approximately 65 cm away, without a chin rest. Our system performed real-time hand gesture recognition via an external webcam, using the Google MediaPipe library³ to process the video stream and extract hand landmarks.

7.3 Interaction Technique

We integrated *ZoomPursuit* into a two-stage multimodal refinement method. The interaction, illustrated in Figure 7, was primarily based on gaze and hand gestures (specifically, a pinch gesture for selection and a raised index finger to trigger correction). The workflow allowed a user to either select a target directly with a pinch gesture or trigger *ZoomPursuit*’s automated correction by raising their index finger. During correction, the cursor was temporarily hidden and reappeared at the new, corrected position. A minor implementation bug resulted in the cursor not being hidden for the first two participants; this was fixed for all subsequent participants. Afterward, the user could manually refine the cursor position with their hand, confirm with a final pinch, or cancel by lowering their finger.

Two underlying mechanisms supported this two-stage refinement method. First, a Recursive Least Squares (RLS) filter⁴ provided dynamic calibration by incrementally learning and correcting systematic pointing errors. Second, for manual refinement, the system set a dynamic step-length for hand movements based on the error magnitude estimated during the *ZoomPursuit* stage. This adaptive refinement was designed to balance speed and accuracy: a large estimated error resulted in a larger step-length for quick adjustments, while a small estimated error led to a smaller step-length, enabling accurate and precise final tuning. We explained these mechanisms to participants before the study.

³https://mediapipe-studio.webapps.google.com/demo/hand_landmarker

⁴Our C# implementation of the RLS filter was adapted from a Python version available at <https://github.com/craig-m-k/Recursive-least-squares/tree/master>

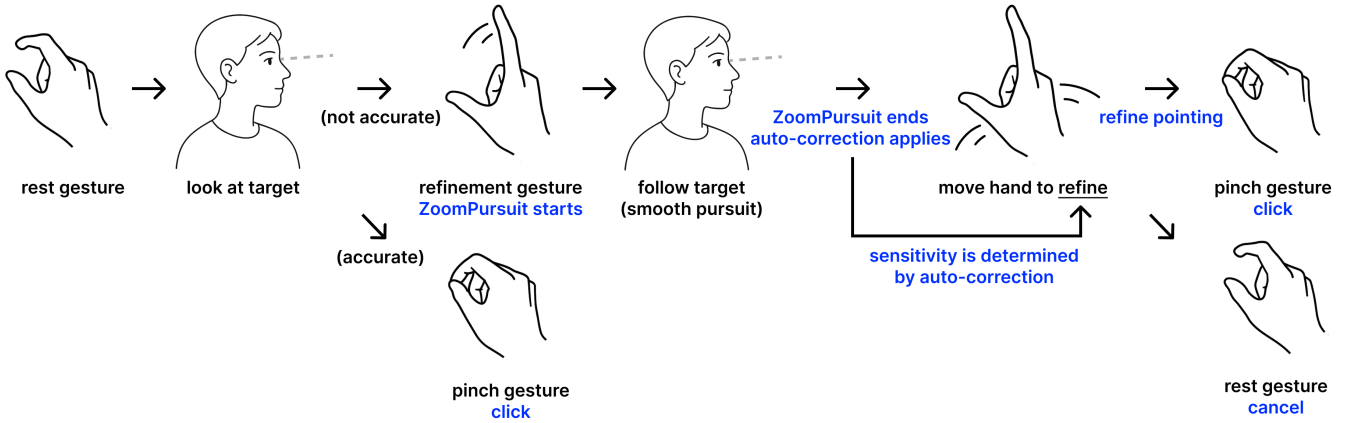


Figure 7: The two-stage interaction workflow. A user first looks at a target. They can then either perform a pinch gesture for direct selection or raise their index finger to trigger the automated ZoomPursuit correction. After the automated correction, the cursor appears at a more accurate position, and the user can perform fine-tuning with hand movements before confirming with a final pinch.

7.3.1 Procedure. The session began with a consent form and a demographic questionnaire (Appendix A.2). Each 40-minute session consisted of four steps:

- (1) **Setup:** After consent and demographics, we demonstrated the system. A baseline accuracy assessment then collected a benchmark set of raw gaze data and measured the pre-task pointing error.
- (2) **Task 1 (WikiGame):** Participants performed the task, which also trained the dynamic calibration model, and then completed a System Usability Scale (SUS) questionnaire.
- (3) **Task 2 (Selection Task):** We cleared the system’s calibration data. This was done to simulate a “cold start” scenario and assess the dynamic calibration model’s performance from an uncalibrated state. Participants then performed the 20-target pointing task, followed by a NASA-TLX questionnaire.
- (4) **Final Steps:** We performed a standard Tobii 5-point calibration, immediately followed by a final accuracy assessment. The session concluded with a semi-structured interview (protocol in Appendix A.3.2).

7.4 Data Analysis Strategy

We conducted a mixed-methods analysis. Quantitative log data addressed system performance (**RQ2**, **RQ3**, **RQ4**), while qualitative interview data addressed user experience (**RQ5**). We analyzed data from the WikiGame and the pointing task separately.

7.4.1 Quantitative Analysis. To answer **RQ2** (Automated Correction), we used a Linear Mixed-Effects Model (LMM) to evaluate the relationship between initial pointing error and the compensation applied by ZoomPursuit, with a random intercept per participant. To answer **RQ3** (Overall Performance) and **RQ4** (Dynamic Calibration), we analyzed two primary metrics. First, we measured the final pointing accuracy in the Selection Task after both correction

stages (for **RQ3**). Second, to evaluate the effectiveness of the dynamic calibration component (for **RQ4**), we compared the pointing accuracy of two states: 1) the final mean error by applying our fully trained dynamic model to the raw gaze data from the baseline assessment, and 2) the mean error from the same assessment after a standard Tobii calibration.

Data Preprocessing and Trial Exclusion. We excluded trials that were clear mis-activations, such as those caused by gesture recognition errors, based on quantitative criteria (e.g., final pointing error far exceeding the target radius).

7.4.2 Qualitative Analysis. To answer **RQ5** (User Experience), we transcribed the semi-structured interviews and used Thematic Analysis to identify key themes regarding the two-stage refinement method.

8 Study 2: Follow-up Study Results

This section reports the quantitative and qualitative results of our evaluation, addressing the model’s automated correction performance (**RQ2**), its overall pointing accuracy (**RQ3**), its on-the-fly calibration capability (**RQ4**), and the user experience (**RQ5**).

8.1 Automated Correction Performance

To evaluate the automated correction mechanism, our first step was to answer **RQ2: How accurate is the ZoomPursuit correction (first stage)?** We used a Linear Mixed-Effects Model (LMM) to analyze the relationship between initial pointing error and the compensation from ZoomPursuit (Figure 8). For this analysis, we filtered the interaction data, excluding trials where participants skipped the automated correction and trials with a final pointing error greater than 0.5° . We filtered the latter because large final errors indicated a missed target or a gesture recognition error. This filtering resulted in 125 valid trials for analysis.

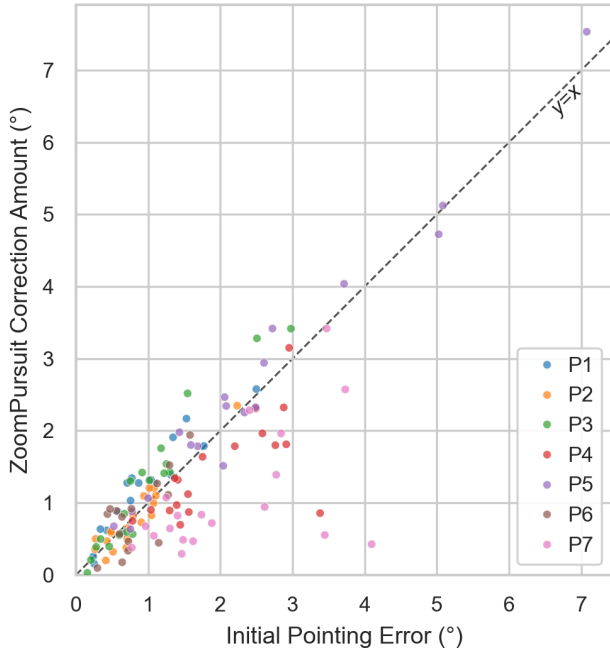


Figure 8: The relationship between initial pointing error and the system’s automated correction amount for each trial in the pointing task of the follow-up study (Study 2). The strong positive correlation (slope ≈ 0.88) demonstrates that the automated correction mechanism reliably compensated for a large proportion of the initial error. Each participant’s data is shown in a different color.

The LMM on all 125 trials showed a significant positive relationship, with the automated correction accounting for 88.3% of the initial error ($\beta = 0.883$, $p < 0.001$, $R_m^2 = 0.703$). A sensitivity analysis was conducted excluding data from one participant (P7). Interview data revealed this user frequently failed to adhere to the interaction protocol by not maintaining gaze on the target during the pursuit phase, thus violating the technique’s core assumption. Excluding this outlier’s 19 trials (leaving $N=106$ for analysis) showed the slope increased to 0.985 ($p < 0.001$), the marginal R^2 increased to 0.853, and the mean reduction in error magnitude increased from 62.2% to 70.3%. This indicates the automated correction functioned as expected for most users.

A check of LMM assumptions found that while random effects were normally distributed (Shapiro-Wilk, $p > 0.05$), the model’s residuals were not ($p < 0.001$). Given that LMMs are robust to violations of this assumption and a residuals vs. fitted plot showed homoscedasticity, we proceeded with the analysis.

8.2 Overall Pointing Performance

After assessing the first stage, we evaluated the overall performance of the complete two-stage refinement method. This involved analyzing the final pointing accuracy (RQ3) and the system’s ability to calibrate over time (RQ4).

8.2.1 Final Pointing Accuracy. To answer **RQ3: How accurate is the two-stage design?**, we measured the final pointing error after both automated correction and manual refinement. The two-stage mechanism was highly effective at improving accuracy within a single interaction. In the Selection Task, the initial mean error of 1.47° was first reduced to 0.56° by the automated ZoomPursuit correction. It was then further reduced to a final mean error of 0.10° ($SD = 0.08$), or approximately 5 pixels on the display, after manual refinement.

8.2.2 Dynamic Calibration Improvement Over Time. Next, we addressed **RQ4: Can the two-stage refinement technique support on-the-fly calibration during interaction?** To do this, we analyzed how pointing error changed as participants progressed through the tasks (Figure 9). We used a Linear Mixed-Effects Model (LMM) for each task to assess the trend.

For the Selection Task, the LMM showed a significant decrease in pointing error with each subsequent operation ($\beta = -0.0864$, $p < 0.001$). The model indicates a 76.7% reduction in pointing error over the 20-target task, demonstrating a clear calibration effect as the system gathered more data. Our analysis of the baseline accuracy assessment data further supported this. The initial raw mean error of 1.892° ($SD=0.832$) was reduced to 0.839° ($SD=0.196$) when our final, dynamically trained model was applied. For comparison, a standard Tobii calibration on the same setup yielded a mean error of 0.679° ($SD=0.135$).

For the WikiGame, the LMM also showed a statistically significant decrease in pointing error over time ($\beta = -0.0006$, $p = 0.030$). However, we must interpret this result with caution. Experimenter notes and interaction logs revealed that the gesture recognition implementation had robustness issues, which may have acted as a confounding variable. Once participants adapted to the gesture system, the dynamic calibration appeared to function as intended (Figure 9, top).

8.3 User Experience and Qualitative Findings

Finally, to answer **RQ5: What is the user experience of using the two-stage refinement design?**, we analyzed data from questionnaires and semi-structured interviews.

8.3.1 Questionnaire Results: Usability and Workload. The System Usability Scale (SUS) yielded a mean score of 66.1 ($SD = 14.4$), with individual scores from 47.5 to 87.5.

The NASA-TLX results showed an overall workload with a mean raw score of 27.0 ($SD = 13.1$) out of 100. An analysis of individual dimensions revealed that Effort ($M=43.1$, $SD=29.9$) and Mental Demand ($M=36.0$, $SD=18.2$) were the dimensions contributing most to workload, consistent with qualitative findings. The score for Frustration was low ($M=11.0$, $SD=12.7$).

8.3.2 Qualitative Findings from Interviews. Thematic analysis of interview data revealed five findings about the user experience of the two-stage refinement method.

Finding 1: Cognitive Load Modulated Interaction Effort. The effort required to interact with the two-stage model was not constant; instead, it varied significantly with the cognitive demands of the primary task. Maintaining a steady gaze during the automated

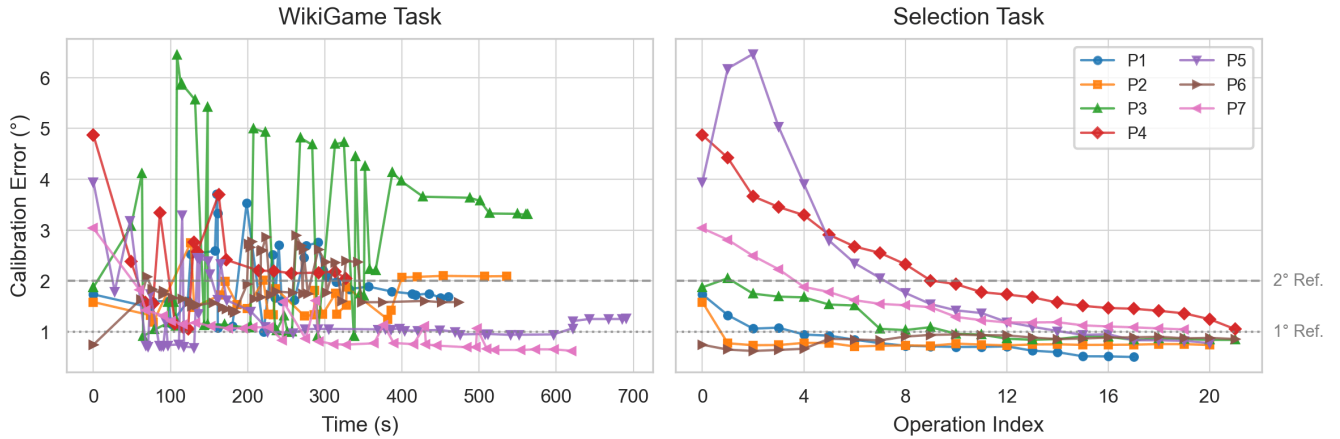


Figure 9: Calibration error over time during the follow-up study (Study 2), for the WikiGame (left) and the Selection Task (right). Each point represents one interaction. The data shows a general decrease in error as the dynamic calibration model gathered more data, particularly in the structured Selection Task. The higher variance in the WikiGame task (left) reflects challenges with the gesture recognition implementation. For example, some users initially struggled with the pinch detection threshold, causing unintentional activations or cancellations that introduced noisy data into the dynamic calibration model. The more consistent trend in the structured Selection Task (right) suggests users had become more familiar with the system’s thresholds, leading to fewer errors.

correction phase felt effortless in the low-load selection task but became a conscious effort in the high-load WikiGame. This conflict was most pronounced in the WikiGame task. Here, the requirement to fixate on a hyperlink directly competed with the natural saccadic eye movements of reading, forcing users to consciously suppress their reading behavior. This created a unique challenge not typically present in standard GUI pointing tasks.

Finding 2: Adaptive Sensitivity Traded Predictability for Accuracy. While the adaptive sensitivity in the manual refinement stage allowed for high precision, this came at the cost of predictability for some users. Most users reported that the fine-grained control for the final step was “very smooth” (P1) and felt confident in their accuracy (P3). However, not all users clearly perceived the changing sensitivity. One participant (P7) found the unpredictable speed “harder to use,” noting that “sometimes I thought I was moving my hand, but it was moving very slowly. Sometimes I’m moving very fast.” Another (P5) correctly deduced the mechanism, realizing the cursor slowed down “once the calibration gets pretty well.” This highlights a design challenge: implicit system adaptation can optimize performance but may reduce the predictability of the user’s mental model.

Finding 3: System Improvement Was Perceived as User Mastery. The system’s dynamic calibration was effective, leading to a palpable improvement in accuracy over time that most participants noticed. Interestingly, they did not always attribute this improvement to the system itself, but rather to their own developing skill. As the system provided more consistent control, users felt they were “calibrating myself” (P2) or that their “ability to... control it with my hand... felt like it was getting easier” (P1). This suggests

the system’s adaptation was well-integrated, leading users to experience a sense of mastery.

Finding 4: A Manageable Learning Curve in Decoupling Eye and Hand Coordination. The primary learning challenge of the two-stage workflow was internalizing the need to decouple eye and hand functions, a process that required overriding strong sensorimotor habits. One participant (P4) articulated this as the need to “task switch between what the eyes do, and then... you have to focus on what the hand does.” This separation was challenging because of the common tendency for eyes to follow hand movements and vice-versa. For instance, one user noticed their eyes moving during the hand-control phase (P5), while another found it hard to stop their hand from “waving in the air during the looking phase” (P4). Internalizing this decoupled interaction model constituted the learning process for the two-stage refinement method.

Finding 5: The Cost of Misalignment: Exploring a Boundary Case. The experience of one participant (P7) highlighted a key boundary condition of the interaction model. This user, who expressed a strong preference for manual control, frequently disregarded the core interaction protocol. Instead of tracking the target during the zoom, they would wait for the automated stage to finish, then fixate on the newly-positioned cursor to visually guide their hand during manual refinement. This behavior—looking at the cursor instead of the target—violated a core assumption of the pursuit-based correction mechanism, as the system received incorrect gaze data for its error estimation. Consequently, the automated correction was less effective for P7 (as shown in Figure 8). This, in turn, led to smaller estimated errors, causing the adaptive manual refinement to use a smaller, less effective step-size, which the user perceived as the system being “harder to use.” This case demonstrates that

the model’s effectiveness is contingent on the user’s adherence to the decoupled interaction paradigm.

9 Discussion

This work addressed a key challenge for smooth pursuit interaction: how to retain its core benefits, such as robustness to absolute error and the potential for calibration-free interaction, while making the technique applicable to general-purpose, unmodified GUIs. Our solution is ZoomPursuit, a technique whose viability we demonstrated through an evaluation of its core mechanism and its integration into a broader two-stage interaction workflow.

9.1 Addressing the Core Challenge: Pursuit Interaction on Unmodified GUIs

By its design, our approach is inherently applicable to unmodified GUIs. It repurposes system-level magnification, a function independent of any specific application’s content, to generate the necessary motion stimulus. This mechanism allows the technique to point at any location on the screen, fulfilling a key design goal.

Our baseline evaluation (Study 1) then established that this core mechanism is effective. Under simulated calibration drift, it achieved robust pointing performance, matching the accuracy of a manual refinement method (*ZoomDPad*) with the speed of a direct selection method (*ZoomCenter*). This validated the fundamental principle of our approach.

The follow-up study demonstrated how the core mechanism could be successfully integrated into a practical, two-stage workflow. In a more challenging, uncalibrated scenario, this complete design achieved a final pointing error of 0.10° and supported on-the-fly dynamic calibration.

9.2 Obtaining High-Fidelity Data for Dynamic Calibration

A key advantage of pursuit-based methods is their potential for on-the-fly calibration, but this hinges on collecting reliable “gaze-target” data pairs during interaction. Our findings suggest that ZoomPursuit’s design is particularly well-suited for this. The action required for its error correction mechanism, which is maintaining gaze on a moving target, is a natural reflection of the user’s intent to select that target. This ensures that the measured gaze shift corresponds to the actual target displacement, producing high-fidelity data.

This contrasts with the other techniques evaluated. With *ZoomCenter*, some users learned to compensate for drift by intentionally looking away from the target, a strategy that would provide a dynamic calibration model with misleading data. With *ZoomDPad*, the constant availability of manual refinement meant users had little incentive to place their initial gaze accurately, rendering the initial fixation an unreliable signal of intent. By requiring an action that aligns user intent with the data collection mechanism, ZoomPursuit establishes a reliable channel for capturing the data needed for dynamic calibration.

9.3 The Trade-off Between Automation and User Agency

In our baseline evaluation, ZoomPursuit’s objective performance under calibration drift did not translate to a better subjective experience for all participants. The underlying issue was that the automated correction was imperceptible, leaving users without final control when the automation was not perfectly accurate. The two-stage refinement method in our follow-up study resolved this by combining automated assistance with manual final control.

This resolution, however, introduces a trade-off. A key finding from our baseline evaluation highlights this: under accurate calibration, the manual *ZoomDPad* was significantly more accurate than *ZoomPursuit*. This suggests a user need for deterministic final control when errors are low. The broader implication is a design trade-off where a single-step interaction is sacrificed for robustness, accurate and precise control, and user agency. The value of this trade-off is context-dependent.

9.4 Limitations

9.4.1 User Studies. Our findings are based on a non-diverse population of students and researchers, which may limit the generalizability of our results to a broader population. Additionally, the two studies used different eye trackers and display setups. This hardware difference is a confounding variable, so we treat the two studies as sequential but separate investigations, with no direct quantitative comparisons made between them.

9.4.2 Experimental Manipulations. Our simulation of calibration drift warrants discussion. While amplifying each user’s innate error field does not perfectly replicate all forms of natural drift, it provided a methodological advantage: a controlled and replicable way to create two statistically distinct error conditions. This controlled approach was essential for isolating the effect of error magnitude on performance.

9.4.3 Implementation. A key limitation was the brittleness of the hand gesture recognition in the follow-up study. The experience of one participant (P7) highlighted a related boundary condition. This user did not adhere to the interaction protocol of tracking the target, which degraded the system’s performance. This case demonstrates that the model’s effectiveness is contingent on the user’s adherence to the decoupled interaction paradigm. Furthermore, the current implementation of the zoom mechanism does not account for screen boundaries. When a target is near the edge of the screen, the magnification can cause parts of the view to be cropped. While this can be addressed by shifting the viewport, it was not implemented in the current work.

The findings on dynamic calibration are based on a small sample ($N=7$) and should be interpreted as a proof-of-concept. The consistent calibration effect observed across most participants, however, suggests that larger-scale investigation is warranted. Our finding that initial error did not predict final error for *ZoomPursuit* (contradicting H4) also warrants further investigation.

9.5 Future Work

9.5.1 Accessibility. One potential application for this work is in accessibility for users with motor disabilities. For a user with a

personalized calibration, the error field from postural drift should be uniform and locally smooth (a minimal ΔE term in our model). In this scenario, a single gaze-only ZoomPursuit interaction could correct the offset. This could change calibration drift from a recurring interruption into a transient issue resolved through normal use.

9.5.2 Improving Low-Cost Eye Tracking. Future research could focus on low-cost eye trackers. The primary barrier for these devices is often their poor absolute accuracy. Because ZoomPursuit's error correction is based on relative motion, it may offer a path to significantly improve the performance and viability of these highly accessible devices.

9.5.3 Opportunities for Adaptive Interaction. ZoomPursuit opens an opportunity for adaptive interaction. Because the estimated error E is reliable, future systems could use it not just to correct gaze coordinates, but to adapt the interaction model in real time. For example, the magnitude of E could be used to dynamically set various interaction parameters, going beyond the adaptive step-length used in our second study. One such parameter could be the control activation radius of a ZoomDPad-like manual refinement stage. If this radius (the "safe" area around the center) is smaller than the current pointing error E , a user looking at the target might unintentionally trigger a directional input. Conversely, an overly large radius would require fatiguing eye movements to activate the controls. The real-time error E provided by ZoomPursuit could thus be used to set an optimal activation radius on the fly, balancing safety and comfort. This would create a system that adapts its core mechanics based on a reliable, real-time measure of pointing error.

9.5.4 New Contexts and Modalities. The principles of ZoomPursuit could be extended to VR and MR environments. As 2D user interfaces are likely to remain a prevalent interaction paradigm in these immersive contexts, our findings can be directly applied to enhance gaze-based interaction on such interfaces. For 3D interaction, the core mechanism could also be adapted: inducing motion through magnification and tracking the 2D retinal displacement of 3D objects could enable accurate selection at a distance. Future work could also explore different second-stage modalities in the VR and MR contexts.

10 Conclusion

This paper presented ZoomPursuit, a technique that extends smooth pursuit interaction to unmodified general-purpose GUIs. Through two sequential studies, we demonstrated its ability to achieve a final pointing error of 0.10° and to calibrate on the fly from an uncalibrated state. Our investigation also highlighted a conflict between automation and user agency, which led to a two-stage multimodal design.

This work offers two contributions. First, it presents and evaluates ZoomPursuit, a technique that provides a viable model for robust, calibration-free gaze pointing on unmodified GUIs. Second, it presents a two-study evaluation that analyzes the conflict between automation and user agency, resulting in a practical pattern for balancing the two.

Gaze interaction has historically struggled to meet the accuracy demands of GUIs designed for mouse input. Instead of trying to

force gaze into the role of an accurate and precise pointer, our work provides a model that adapts the interaction to the user's natural abilities. By inducing and analyzing a smooth pursuit movement, this approach leverages what the eye does well to correct for inherent pointing inaccuracies. This approach thus provides a model for adapting smooth pursuit-based interaction to the specific demands of unmodified, general-purpose GUIs.

Ethical Considerations

The Ethics and Privacy Quick Scan of the Utrecht University Research Institute of Information and Computing Sciences classified this research as low-risk with no fuller ethics review or privacy assessment required.

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I want to thank my supervisor, Lynda Hardman, for her support and guidance throughout this project. I greatly benefited from our meetings, which helped shape the direction of this work. I also want to thank my examiner, Wolfgang Hürst. His insightful courses on Multimodal Interaction and Mobile Interaction are the foundation of this research.

Author's Note on the Use of Generative AI

I acknowledge the use of generative AI as a supportive tool in preparing this thesis. It assisted with several tasks, including: proof-reading and refining language, phrasing complex arguments, formatting LaTeX source code, typesetting equations, and creating the hand gesture sketches and the avatar in the figures. For the code implementation and data analysis, it helped with debugging code, looking up specifications for libraries and APIs, and exploring potential implementation strategies.

The AI's role was strictly assistive. It was not used to generate any unsubstantiated scientific content. I retain full responsibility for the integrity of this research and its final conclusions.

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A Appendix

This appendix contains supplementary materials for our user studies.

A.1 Questionnaires Used in the Experiments

A.1.1 NASA-TLX Task Load Index. After each experimental condition, participants completed the NASA-TLX questionnaire to assess subjective task load. The scale consists of six dimensions, which participants rated from 0 to 100.

- **Mental Demand:** How mentally demanding was the task?
- **Physical Demand:** How physically demanding was the task?
- **Temporal Demand:** How hurried or rushed was the pace of the task?
- **Effort:** How hard did you have to work to accomplish your level of performance?
- **Performance:** How successful were you in accomplishing what you were asked to do? (Note: A lower score indicates better performance.)
- **Frustration:** How insecure, discouraged, irritated, stressed, and annoyed were you?

A.1.2 User Usability Questionnaire (UMUX-Lite). After testing each interaction technique, participants completed the UMUX-Lite questionnaire to assess its usability. It uses a 5-point Likert scale (1=Strongly disagree, 5=Strongly agree) for two statements:

- This interaction technique helps me accomplish the task effectively.
- This interaction technique is easy to use.

A.2 Demographic and Experience Questionnaire

Before the experiment, participants completed a questionnaire about their demographic information and prior experience with eye-tracking systems. The questions were as follows:

- **What is your age range?**
 - 18-24
 - 25-34
 - 35-44
 - 45-54
 - 55 or above
- **What is your gender?**
 - Male
 - Female
 - Non-binary / third gender
 - Prefer not to say
- **Do you wear glasses or contact lenses?**
 - Yes, glasses

- Yes, contact lenses
- No
- **Is your vision fully corrected when wearing your glasses or contact lenses?**
 - Yes, my vision is normal when corrected.
 - No, I still have some vision difficulties even with correction. (If selected, a text box was provided for details.)
- **Do you experience any other vision-related issues (even when corrected)?**
 - No
 - Yes (If selected, a text box was provided for details.)

A.3 Semi-structured Interview Protocols

After completing all tasks, we conducted semi-structured interviews based on the following protocols.

A.3.1 Study 1 Interview Protocol.

- A. **Opening**
 - (a) “Thank you again for completing all the tasks. Now, I’d like to ask a few questions about your experience. As we discussed earlier, I would like to record our conversation, with your permission. Is that still okay with you?”
- B. **Per-Method Review.** Conducted for each of the three techniques, in the order the participant used them.
 - (a) *(While a video clip of the technique plays on a loop)* “Do you feel this technique helped you select points on the screen more precisely?”
 - (b) “Did you find it easy or difficult to learn?”
 - (c) “Could you explain why?”
 - (d) “Did your eyes feel tired while using it?”
 - (e) “Were there moments that felt smooth or frustrating?”
 - (f) “Did you ever need to intentionally look away from the target to use this technique?”
 - (g) “Did you find yourself adjusting your head posture to aim more accurately?”
 - (h) *Zoom Pursuit only:* “Did you find it helpful or distracting?”
- C. **Overall Comparison.**
 - (a) *(While side-by-side video clips of the techniques play on a loop)* “On a scale of 1-10, how would you rate the eye-tracking accuracy in this experiment?”
 - (b) “Please rank the three techniques by your personal preference and briefly explain why.”
- D. **Closing.**
 - (a) “Do you have any other thoughts or suggestions that we haven’t covered?”
 - (b) “Do you have any questions for me?”

A.3.2 Study 2 Interview Protocol.

- (1) The method requires you to maintain your gaze on the target throughout the zoom. Did this action feel natural, or did you find the requirement unusual or demanding?
- (2) During the two tasks, did you ever forget or fail to follow this gaze requirement?

- (3) Did you experience any eye strain or fatigue while using the system? If so, what do you think was the main cause?
- (4) Did you feel this method enabled you to accurately select very small targets?
- (5) The system’s calibration was reset before each task. During each task, did you perceive the system’s accuracy improving over time? For example, did the pointer seem to become more stable or easier to control?
- (6) What was your experience with the manual refinement stage? Did you find it easy or difficult to use your hand for fine-tuning?
- (7) Do you have any other suggestions for the overall interaction method?

A.4 Full LMM Results from Study 1

The following table provides the complete output from the Linear Mixed-Effects Models (LMMs) used to analyze data from Study 1.

Table 1: Linear Mixed-Effects Model (LMM) results for all dependent variables. The reference levels are *Drifted* for the group factor and *ZoomPursuit* for the technique factor. Pointing error was square-root transformed and task time was log-transformed. Significant p-values ($p < .05$) are marked in bold. For subjective ratings, p-values are FDR-corrected.

Predictor	Pointing Error (sqrt)		Task Time (log)		Workload (NASA-TLX)		Usability (UMUX-Lite)	
	Coefficient (β)	p-value	Coefficient (β)	p-value	Coefficient (β)	p-value (corr.)	Coefficient (β)	p-value (corr.)
Intercept	0.537	<.001	1.464	<.001	48.60	<.001	53.57	<.001
Group [Accurate]	0.006	.935	0.064	.667	-1.19	.880	16.07	.309
Technique								
ZoomCenter	0.261	<.001	-0.048	.421	-14.41	.070	19.64	.118
ZoomDPad	-0.029	.393	0.643	<.001	-19.60	.020	21.43	.106
Group $\times$ Technique Interaction								
Accurate $\times$ ZoomCenter	-0.311	<.001	0.070	.404	16.83	.106	-14.29	.385
Accurate $\times$ ZoomDPad	-0.073	.127	-0.363	<.001	22.38	.059	-26.79	.118
Marginal R^2 (R_m^2)	0.121		0.156		—		—	
Conditional R^2 (R_c^2)	0.253		0.329		—		—	